

HW-TSC’s Submissions To the IWSLT 2026 Offline Speech Translation Task

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Abstract

This paper describes the HW-TSC’s submission to the IWSLT 2026 Offline Speech Translation Task, specifically for the English-to-Chinese and English-to-German unconstrained tracks. Our system adopts a robust cascade architecture optimized for long-form, unsegmented audio. To mitigate the hallucination and inconsistency issues common in long-sequence processing, we propose a two-pass transcription strategy: an initial streaming ASR with a 12-second context buffer for sentence-level coherence, followed by Qwen3-ForcedAligner for precise timestamping. Based on these alignments, a second-pass refinement is conducted using Qwen3-Omni on re-segmented 30-second chunks to ensure high-fidelity transcriptions. For the translation module, we employ a context-aware segment merging strategy (up to 150 tokens) to empower the Qwen3 llm with sufficient semantic context. Experimental results on the tst-2022 benchmark demonstrate the effectiveness of our pipeline, achieving COMET scores of 0.8462 (En-Zh) and 0.7854 (En-De), significantly outperforming the standard cascade baselines.

1 Introduction

The rapid evolution of multimodal large language models (MLLMs) (Yin et al., 2024) and end-to-end speech processing (Bérard et al., 2016) has inaugurated a new era of efficient and robust speech translation. As a pivotal cross-modal task, speech translation bridges linguistic gaps and enables seamless communication in scenarios such as international conferences and media localization. The IWSLT 2026 Offline Unconstrained Track serves as a rigorous benchmark to evaluate systems under realistic conditions. It encourages the use of state-of-the-art pretrained models and advanced engineering to address challenges such as long-form audio and noisy environments.

While end-to-end models such as Whisper (Radford et al., 2023) and SeamlessM4T (Barrault et al., 2023) have made significant progress, they still encounter limitations in data scalability and robustness within unconstrained scenarios, particularly regarding hallucination control in zero-shot settings. In contrast, cascade speech translation systems (Bentivogli et al., 2021) decompose the task into automatic speech recognition (ASR) (Gulati et al., 2020) and machine translation (MT) (Vaswani et al., 2017). This paradigm offers superior flexibility, interpretability, and compatibility with mature pretrained models. Recent studies have further demonstrated that leveraging Large Language Models (LLMs) as translation engines can significantly improve semantic consistency compared to traditional NMT systems (Zhu et al., 2024). Considering the complexity of the IWSLT 2026 test set, we adopt a cascade architecture for our submission, focusing on English-to-Chinese and English-to-German directions.

We leverage the unconstrained track settings to design a robust pipeline centered on the Qwen3 (Yang et al., 2025) model family. First, we apply voice activity detection (VAD) using the Silero VAD framework (Team, 2024) to filter non-speech segments, which reduces computational overhead and mitigates transcription hallucinations. For the ASR module, we implement a streaming inference framework featuring context caching and a two-pass transcription strategy. We utilize Qwen3-Omni (Xu et al., 2025) and Qwen3-4B for initial transcription and employ Qwen3-ForcedAligner (Team, 2026) to generate precise timestamps. Subsequently, we re-segment the audio based on these timestamps for a second-pass refinement to enhance recognition quality. Finally, we merge the refined transcriptions into semantic segments and feed them into the Qwen3-8B for translation, employing a context-aware prompting strategy to maximize translation fidelity (Jean and Cho, 2019).

The remainder of this paper is organized as follows. Section 2 describes the datasets used in our experiments. Section 3 details the system architecture, including audio preprocessing, streaming ASR, and the translation module. Section 4 presents the experimental setup and evaluation results. Finally, Section 5 summarizes our findings and outlines future research directions.

2 Dataset

2.1 ASR Data and Adapter Tuning

We utilize six large-scale speech datasets to train the adapter and fine-tune the decoding capabilities of our ASR module: MuST-C V2 (Cattoni et al., 2021), LibriSpeech (Panayotov et al., 2015), TED-LIUM 3 (Hernandez et al., 2018), CoVoST 2 (Wang et al., 2020), VoxPopuli (Wang et al., 2021), and Europarl-ST (Iranzo-Sánchez et al., 2020) (Table 1).

Dataset	Duration (Hours)
LibriSpeech	960
MuST-C V2	590
CoVoST 2	1,802
TED-LIUM 3	453
Europarl-ST	161
VoxPopuli	1,270
Total	5,236

Table 1: Statistics of the speech corpora used for ASR training.

Unlike standard ASR training, we focus on jointly optimizing the speech-to-text alignment and the LLM’s sequence generation. We leverage these datasets to train a multimodal adapter that bridges the Qwen3-Omni speech encoder and the Qwen3-4B language model. Specifically, we supervised the model to predict autoregressive transcriptions where the newline character `\n` is explicitly inserted at the end of complete semantic units. This ensures that the joint model can function as a sentence-aware transcriber, providing semantically coherent segments for the initial transcription phase. To enhance robustness for long-form speech, we also apply the data augmentation method from (?), concatenating adjacent audio clips to simulate continuous, unsegmented input during training.

2.2 MT Data and Quality Control

For the machine translation module, we leverage a massive-scale web-crawled parallel corpus to provide robust linguistic coverage for the unconstrained track. To ensure the reliability of the train-

ing data, we implement a multi-stage quality control pipeline. We first apply a series of heuristic filters to remove noise, including language identification, length-ratio constraints, and character-level deduplication.

Subsequently, we employ a semantic alignment filtering strategy to calculate cross-lingual similarity between sentence pairs. By setting a high similarity threshold, we effectively filter out low-confidence and misaligned pairs, resulting in a high-quality bilingual dataset for the English-to-Chinese and English-to-German directions. This large-scale pre-training is further complemented by fine-tuning on officially provided task-specific data, enabling the Qwen3-8B LLM to better capture the nuances of professional discourse in the IWSLT 2026 test set.

3 Offline Speech Translation System

Our offline speech translation system follows a robust cascade architecture, as illustrated in Figure 1. The pipeline integrates audio preprocessing, streaming ASR with context caching, forced alignment, two-pass transcription optimization, and segment-level machine translation. This modular design is specifically engineered to handle the complexities of long-form, unsegmented audio by ensuring both acoustic precision and semantic coherence.

3.1 Audio Preprocessing

To improve inference efficiency and mitigate transcription hallucinations, we first perform VAD on the input waveforms. We filter out silent segments and non-speech parts, passing only valid speech segments to the subsequent modules. This step effectively reduces redundant computation and prevents unnecessary generation on meaningless audio fragments, establishing a foundation for stable and efficient transcription.

3.2 Streaming Inference and Context Caching

Since the IWSLT 2026 test set does not provide pre-defined segmentation, we must partition the long-form audio ourselves. Given that long-form speech exhibits strong contextual dependencies, maintaining semantic integrity and cross-sentence coherence is crucial for accurate transcription and translation.

To balance recognition consistency with processing efficiency, we implement a streaming inference

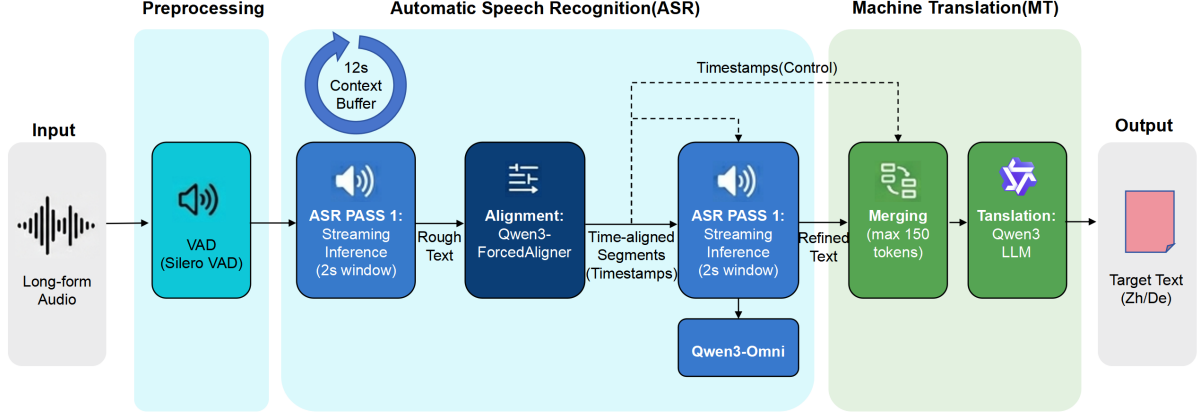


Figure 1: The overall architecture of the HW-TSC offline speech translation system. The pipeline integrates Silero VAD, two-pass ASR (streaming inference and refinement via Qwen3-Omni), Qwen3-ForcedAligner, and context-aware machine translation.

framework with a sliding window mechanism. We process input audio sequentially in 2-second units. To maintain continuity across blocks, we maintain a dynamic buffer that stores historical audio and corresponding text prefixes, with a maximum context length of approximately 12 seconds.

To ensure sentence-level integrity and avoid mid-sentence truncation, we employ a sentence-aware segmentation strategy. We trigger segmentation and output only when the model generates a newline character or reaches the end of an audio block. This design ensures that each output segment constitutes a complete and semantically coherent sentence.

3.3 First-Pass Transcription

In the first-pass transcription, we adopt a joint decoding architecture that integrates speech representations with LLM inference. We utilize a speech encoder derived from the Qwen3-Omni model to extract high-level acoustic features. The text backbone is based on the Qwen3-4B model, which is responsible for generating autoregressive transcriptions. This model is specifically trained to output newline characters at appropriate semantic pauses, ensuring the structural integrity of the segments.

We map speech features into the token embedding space of the LLM via an adapter module. Acoustic representations are injected into the language model input through the `prompt_embeds` mechanism, achieving direct fusion of multimodal information. This structure generates semantically complete initial transcriptions, providing a reliable basis for subsequent timestamp alignment.

3.4 Forced Alignment

To obtain precise temporal boundaries for the transcribed sentences, we apply the Qwen3-ForcedAligner-0.6B model. We perform fine-grained time alignment between the audio waveform and the first-pass transcription output. This model generates accurate sentence-level timestamps, which facilitate reliable audio re-segmentation for the second-pass transcription phase.

3.5 Second-Pass Transcription

We employ a two-pass transcription strategy to further enhance recognition accuracy and robustness. Based on the timestamps obtained from forced alignment, we merge and re-segment the audio into continuous fragments not exceeding 30 seconds. Each fragment is then fed into Qwen3-Omni for a second round of end-to-end speech recognition. This two-pass process effectively corrects recognition errors—particularly in segments involving heavy accents or fast speech—ultimately producing high-quality transcriptions.

3.6 Segment Merging and Translation

Before translation, we merge the refined transcription segments into semantically complete blocks, restricted to a maximum length of 150 tokens. This strategy maintains contextual coherence while preventing issues associated with excessively long model inputs. Finally, we feed the merged source text into the Qwen3-8B LLM to generate the final translation in the target language, completing the offline speech translation pipeline.

4 Experiments and Results

4.1 Experimental Setup

We evaluated our system on the *tst-2022* benchmark dataset. For the ASR module, we used Word Error Rate (WER) (von Neumann et al., 2025) as the primary metric. For the machine translation and the integrated system, we utilized COMET (Rei et al., 2022) score to measure translation quality. All experiments were conducted on the Qwen3 model family, utilizing Silero VAD for audio pre-processing.

4.2 ASR Performance Analysis

To establish a comprehensive understanding of our system’s capabilities, we conduct the ASR evaluation in two stages: model selection under ideal conditions and system validation under realistic unconstrained conditions. All experiments are conducted on the *tst-2022* English-to-Chinese (En-Zh) dataset.

4.2.1 Model Selection and Contextual Capacity

We first evaluate the transcription performance of various models within the Qwen3 family. Since the *tst-2022* dataset provides official segmentation, we use these ground-truth boundaries to establish a performance upper bound. To investigate the models’ ability to leverage long-range dependencies, we further test a strategy where official segments are merged into longer audio chunks not exceeding 30 seconds. The results are summarized in Table 2.

ASR Model	Segmentation Method	WER (%)
Qwen3-ASR-0.6B	Official Seg.	7.31
Qwen3-ASR-0.6B	Official Seg. + 30s	5.79
Qwen3-ASR-1.7B	Official Seg.	5.40
Qwen3-ASR-1.7B	Official Seg. + 30s	3.93
Qwen3-Omni	Official Seg.	5.55
Qwen3-Omni	Official Seg. + 30s	3.14

Table 2: ASR WER (%) on the *tst-2022* test set with official segmentation. Results show that larger models and longer semantic context (up to 30s) significantly enhance recognition accuracy.

The results indicate that **Qwen3-Omni** possesses the strongest acoustic modeling and linguistic reasoning capabilities. Notably, increasing the fragment length to 30 seconds yields a substantial reduction in WER across all models, proving that maintaining semantic integrity via longer segments is crucial for MLLM-based ASR systems.

4.2.2 Evaluation of the Two-pass Strategy

In the realistic IWSLT offline track, official segmentation is unavailable. To verify the effectiveness of our proposed pipeline, we evaluate the system starting from raw, unsegmented waveforms. We compare a **Single-pass** baseline (utilizing streaming ASR with a 12s context buffer but without second-pass refinement) against our full **Two-pass** strategy.

System Configuration	WER (%)
Single-pass (Streaming Baseline)	6.31
Two-pass (Proposed Pipeline)	3.01

Table 3: ASR performance comparison on *tst-2022* without official segmentation. Our two-pass strategy effectively refines initial transcriptions.

As shown in Table 3, our Two-pass strategy achieves a WER of **3.01%**, representing a significant improvement over the single-pass baseline (6.31%). Impressively, this performance even surpasses the best result obtained using official segmentation (3.14% in Table 2). This suggests that our timestamp-based re-segmentation, guided by the Qwen3-ForcedAligner, creates audio boundaries that are more optimized for the model’s receptive field than the original manual annotations. This refined transcription serves as a high-fidelity input for the subsequent translation stage.

4.3 Machine Translation Refinement

To evaluate the effectiveness of our context-aware translation module, we utilize the high-fidelity transcriptions generated by our best-performing **Two-pass ASR system** (3.01% WER) as the input for the translation task. We specifically investigate the impact of our **Segment Merging** strategy by comparing two configurations: a sentence-by-sentence translation baseline and our proposed approach, which merges refined segments into semantic blocks not exceeding 150 tokens.

Configuration	En-Zh	En-De
Qwen3-8B (Sentence-level)	0.8397	0.7628
Qwen3-8B + Merging	0.8462	0.7854

Table 4: Machine translation performance (COMET) on *tst-2022* using Two-pass ASR transcriptions. Segment merging significantly enhances contextual consistency.

As shown in Table 4, the segment merging strategy yields a measurable improvement in COMET scores across both language directions. While the Two-pass ASR provides accurate textual input, translating isolated sentences often leads to a

System	ASR WER ↓		Translation COMET ↑	
	En-Zh	En-De	En-Zh	En-De
Baseline (Single-pass + Std. MT)	6.31	6.54	0.7924	0.7409
HW-TSC (Full Pipeline)	3.01	3.25	0.8462	0.7854

Table 5: Overall system performance on *tst-2022*. The table reflects separate ASR metrics for each language track due to differences in the corresponding audio subsets.

loss of discourse-level information. By grouping refined transcriptions into larger semantic blocks, the **Qwen3-8B** model better captures long-range dependencies and cross-sentence logic. This strategy effectively mitigates the risk of semantic fragmentation, ensuring that the final output maintains the high stylistic and contextual fidelity required for the IWSLT offline track.

4.4 Overall System Evaluation

Finally, we evaluate the end-to-end performance of the integrated HW-TSC system across different language directions. Since the audio subsets for the English-to-Chinese (En-Zh) and English-to-German (En-De) tracks in the *tst-2022* dataset are not identical, we report the ASR Word Error Rate (WER) and translation COMET scores separately for each track. We compare our full cascaded pipeline against a representative baseline system (Single-pass ASR + standard MT).

As shown in Table 5, the integrated HW-TSC system consistently outperforms the baseline across all metrics and subsets. For the En-Zh track, our pipeline reduces the WER from 6.31% to 3.01%, resulting in a significant COMET gain of +0.0538. A similar trend is observed in the En-De track, where the improved transcription fidelity directly contributes to superior translation quality. These results confirm that the synergistic integration of two-pass refinement and context-aware segment merging effectively addresses the challenges of unconstrained offline speech translation, regardless of the target language.

5 Conclusion

In this paper, we presented the HW-TSC team’s submission to the IWSLT 2026 Offline Speech Translation Task. We implemented a robust cascade system centered on the **Qwen3 model family**, specifically optimized for the English-to-Chinese and English-to-German unconstrained tracks. Our core contribution lies in the integration of a **two-pass transcription strategy**—combining streaming inference with context caching and fine-grained

forced alignment—which effectively addresses semantic fragmentation and hallucinations in long-form audio. **Notably, our pipeline demonstrates the ability to surpass official segmentation baselines by optimizing audio boundaries through model-based alignment.**

Experimental results on the *tst-2022* benchmark demonstrate that our pipeline significantly outperforms standard cascade baselines across all metrics. Specifically, the combination of context-aware segment merging and the multimodal capabilities of Qwen3-Omni ensures both high ASR accuracy and superior translation consistency. These findings confirm that a carefully optimized cascade architecture, leveraged with state-of-the-art **Multimodal Large Language Models (MLLMs)**, remains a powerful and flexible paradigm for complex, real-world speech translation scenarios. For future work, we aim to explore **knowledge distillation techniques** to integrate our two-pass insights into more efficient end-to-end frameworks, further reducing inference latency while maintaining the robust performance of our current pipeline.

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